

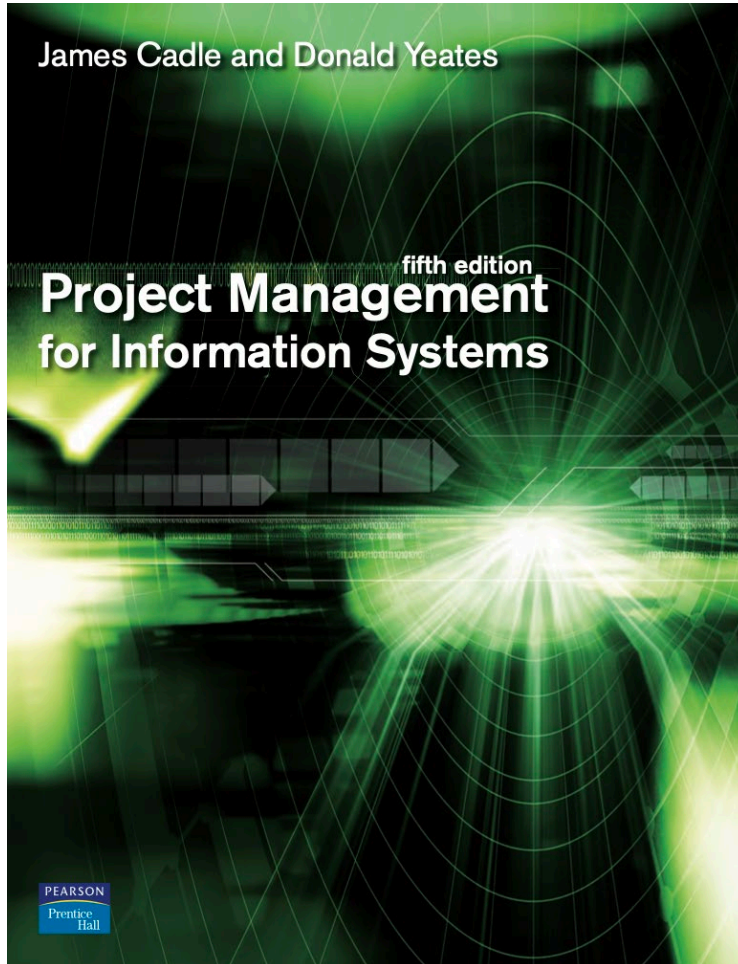
BU5300 – IT Project Management

Lecture 5: Project Profile and Project Planning

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Overview of the Course

Lectures and Book Chapters



Lecture 1 - Chapters 1 and 2

Lecture 2 - Chapters 3 and 4

Lecture 3 - Chapter 6

Lecture 4 - *Agile Development*

Lecture 5 - Chapters 7 and 8

Lecture 6 - Chapter 9 and 10

Lecture 7 - Chapters 11, 12 and 13

Lecture 8 - Chapters 14 and 15

Lecture 9 - Chapters 18 and 21

Lecture 10 - Chapters 20, 22 and 23

IMPORTANT! Changes may occur. Always check on Canvas and TimeEdit

Agenda for today's lecture

- Quick review of Lecture 4
- Chapter 7: Project Profile
 - Project Initiation Document or Project Charter
- Chapter 8: Project Planning, understanding the work

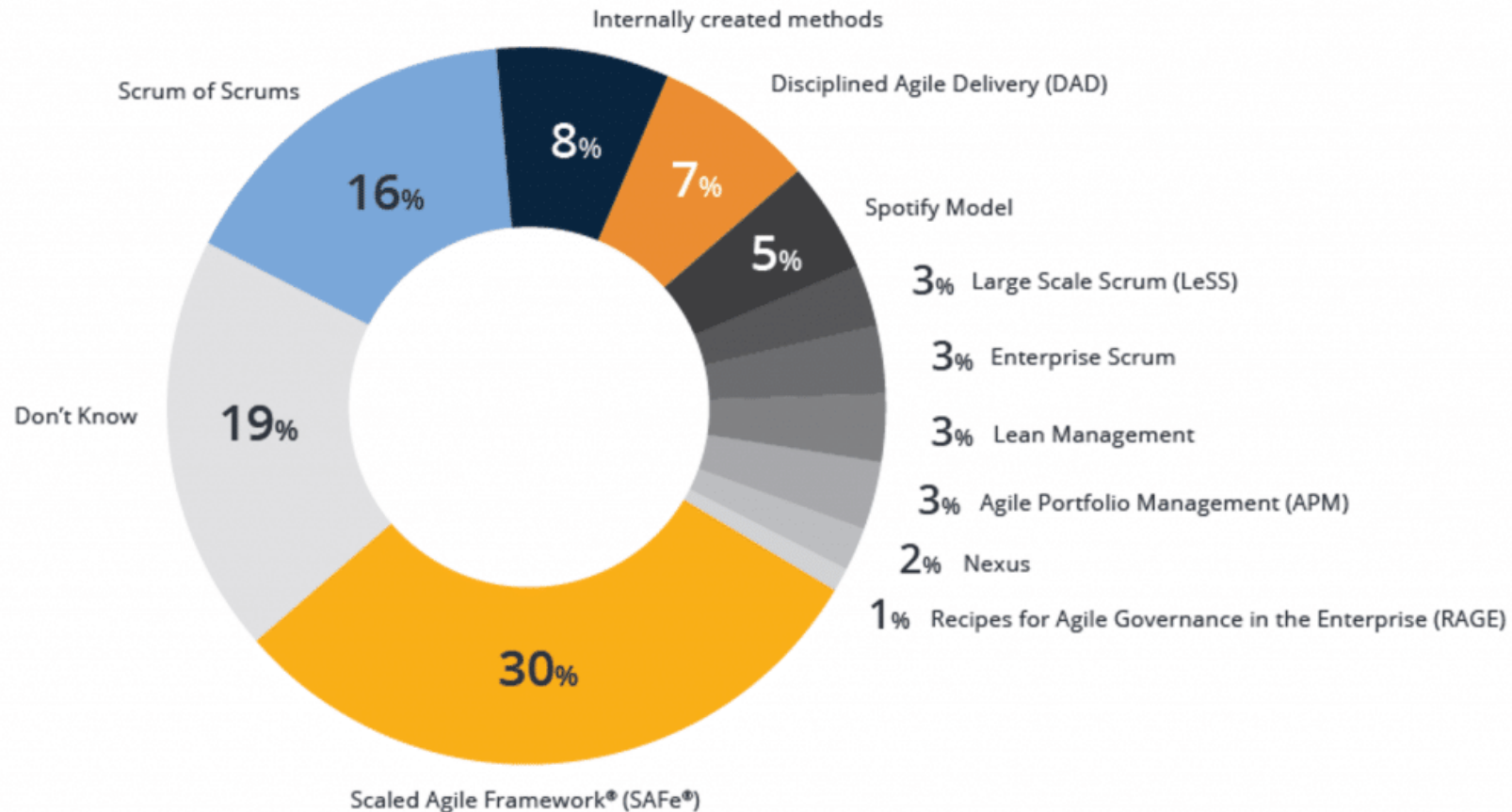
Takeaways from Lecture 4

- Agile is about
 - Individuals and interactions
 - A working product
 - Customer collaboration
 - Responding to changes
- Research has shown that agile is related to successful projects, especially in small-size teams
- Larger agile teams are possible, but there are many challenges

Scaling Agile

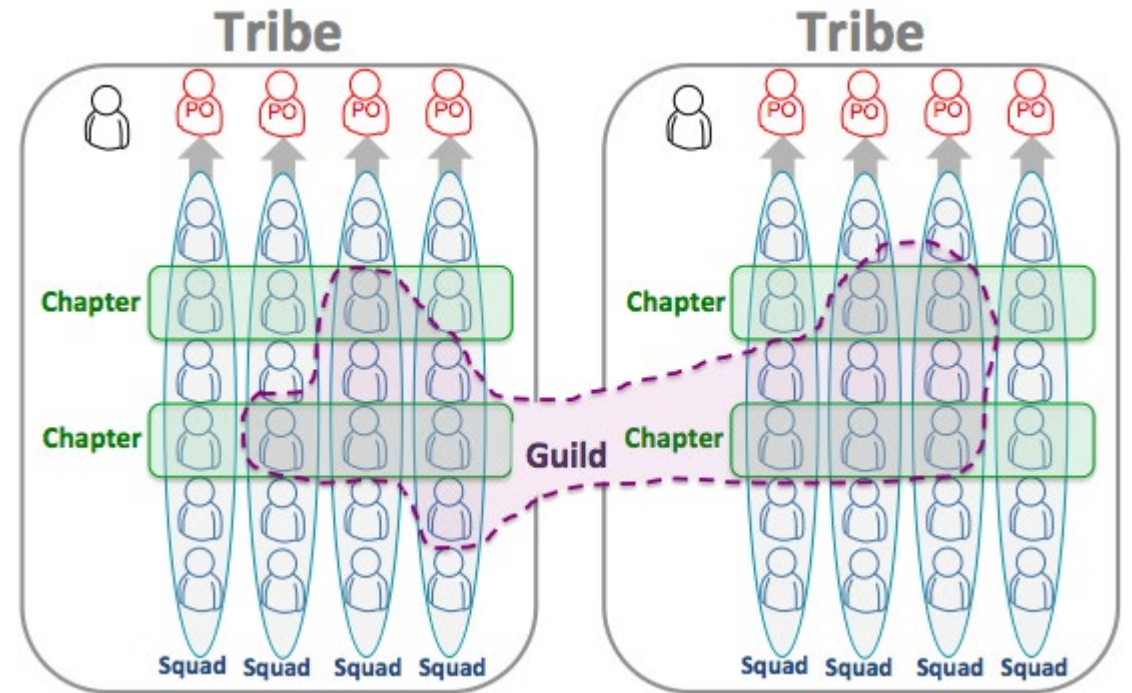
12th Annual State of Agile survey (2018)

The Scaled Agile Framework® continues to be the most popular scaling method cited by respondents (30% this year compared to 29% last year).



Spotify's Model

- **Squads** are like Scrum teams
- **Tribes** are organized by product or project, often in the same place
- **Chapters** are skill communities
- **Guilds** are organized by common interests
- Criticisms:
 - Not Agile!
 - Even Spotify doesn't use it



Chapter 7: Profile of a Project

Learning Outcomes

- Explain the stages of the generic process model
- Explore the importance of the project start-up stage under the headings of what, why, who, how and when
- Prepare a project initiation document (PID)
- Describe the main activities of the development stage

Introduction to the Chapter

- This chapter describes a framework and a structure for what goes on within a typical development project
- An assumption in this chapter:
 - an external supplier will carry out the technical development work alongside the commissioning organization's users
- We shall use 'customer' to mean the organization commissioning the work and 'supplier' to mean the external company brought in to carry out the development work

Project Process Groups (stages)

From the PMI Body of Knowledge



- The 5 phases (stages) of a traditional project (the predictive approach)
- Apply to all types of projects
- Information Systems projects have their own methodologies with their own phases

Project Startup

- Finalize scope – agree on Project Initiation Document (PID):
 - **Objectives:** what the project is about
 - **Scope:** what is in (in scope) and what's out (out of scope)
 - **Constraints:** timescale, budget, methods etc
 - **Authority:** who represents the customer?
 - **Resources:** people, equipment, finance etc
- Agree on roles and responsibilities
- Includes the What, Why, Who, How and When
- Develop initial plans
- Aligns with the “Initiation Phase” In PMI PMBOK and PRINCE2

Products (Deliverables) of the Startup Phase

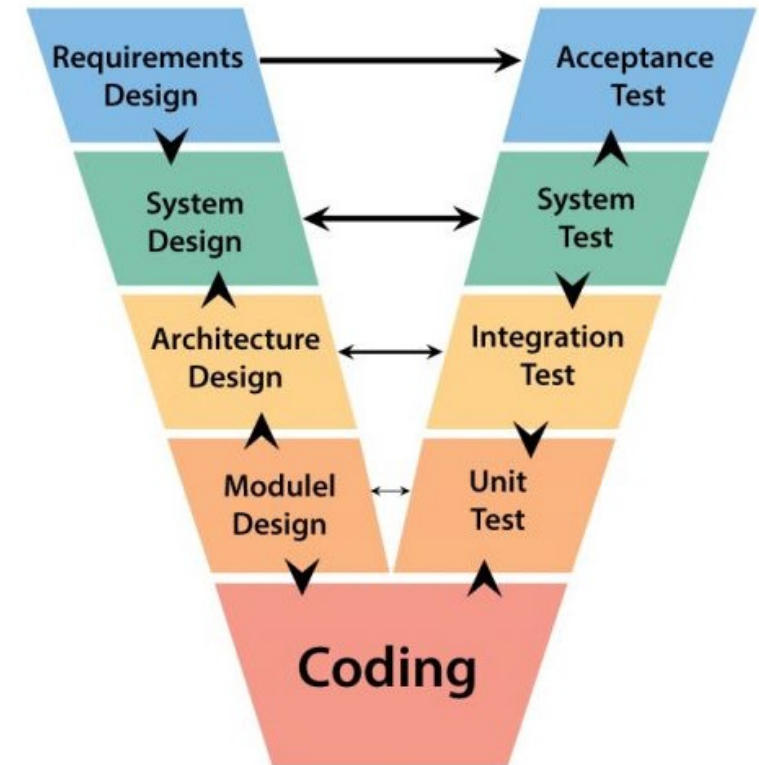
The main products resulting from the Start-up stage are as follows:

- **Project initiation document (PID)**
- **Project plan / charter**
- Quality plan
- Risk management plan
- Project organization structure
- Project administrative procedures

Development stage

- Requirements definition
- Design (*how* the requirements of the project are to be met)
- Implementation (doing the development work)
- Integration and testing
- System testing

See development lifecycles (textbook Chapter 6)



Products (Deliverables) of the Development Stage

The main products resulting from the development stage are as follows:

- Requirements specification
- Technical specification
- Module specifications
- Prototypes
- Completed and tested hardware and software modules
- Acceptable systems test results
- Factory acceptance certificate

Completion stage

- Delivery to the customer
- Training and familiarization
- Acceptance testing
- Customer acceptance
- System commissioning (Go Live)
- Final customer takeover

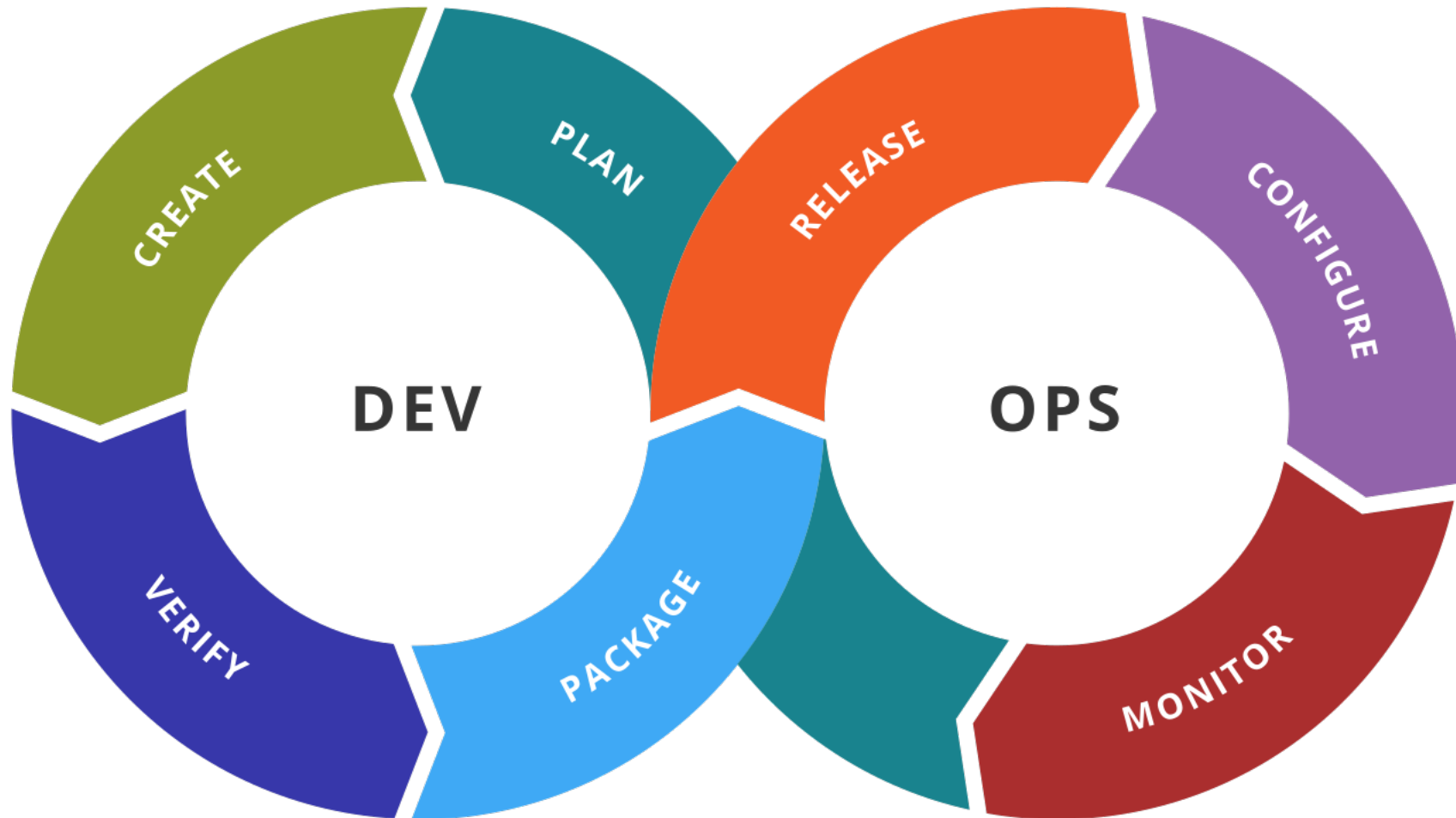
Deliverables (main products resulting from the completion stage) are:

- Site acceptance certificate
- Trained staff
- Commissioned system

Operational stage

- Live running of system
- Fixing problems not found in tests
- Enhancements and modifications
- Products from this stage may include fixes and enhancements

Sidenote: DevOps



Post-project review

- Review should consider:
 - Technical methods and standards: how effective, problems and resolution
 - Project risks: how handled
 - Contractual issues: how resolved
 - Customer/supplier relationship issues
 - Stakeholder management issues
 - Team resourcing issues
 - Project performance against plans
- Aim: to learn from experience and improve management of later projects

PID – Project Initiation Document

Project initiation document (PID)

- Objectives:
 - Business Objectives (from the business case)
 - Project Objectives (what the project itself will deliver)
- Scope: boundaries, activities, deliverables
- Constraints
- Authority
- Resources (people, money, equipment)

Project Initiation

PRINCE and PMBOK

- The process under which the project is created and defined **to the extent necessary to begin planning**
 - It puts the project on a solid foundation
 - Ensures that the organization is clear about the purposes and priorities
 - A place from which the project manager and project board can assess progress
 - Its importance is easily underestimated
- PID (also Project Charter) is the document which outlines the organization's vision for the project and the business case for it

PMBOK vs PRINCE

A few words on PM methodologies

PMBOK (Project Management Base of Knowledge)

- A guide for all-types of project management
- Commonly accepted as best practice for project management
- Comprises a collection of processes and knowledge areas
- Published by the PMI - Project Management Institute (US)

PRINCE (Projects in Controlled Environments)

- A project management methodology
- It is suitable for any size of project
- It is centered on what is to be done and when, with clear and concise steps to follow
- Comprises a set of successive processes that focus on the 'nitty-gritty' of a project
- Sponsored by the UK government

Practical Tips

- There is no single prescribed format
- Continuous evolving is better than a perfect start
 - Update and revise as needed during the project
 - Communicate changes and get people involved
- For larger projects, it is better to divide components in separate documents, i.e. a collection of items within a project management tool
- MS Project helps with that, but even Google Docs, Trello and Wiki would work

Documentation and templates:

<https://kristiania.instructure.com/courses/14965/files/folder/PRINCE2>

<https://kristiania.instructure.com/courses/14965/files/folder/PMBOK>

Project Definition / Scope

Components of a PID

- **What is the project?**
- List its objectives, goals and scope
 - Also the major steps to achieve the objectives
- Identifies the vision for what the project will accomplish
 - Limit the scope of the project as envisioned **when the project is created**
- It is a good idea to **identify the key deliverables**, but not all deliverables
 - Focus on the non-negotiable ones (or the minimum viable product)
 - Other deliverable would be added as the design/planning phase starts

Project Approach / Milestones

Components of a PID

- **How is the project carried out?**
- The processes and procedures that will be used to accomplish the objectives
- Also, list **major stages/events** in the project development
 - A description of the key milestones can aid in the communicating the project to the organization
- Should **NOT** contain a detailed schedule at this point!

Business Case and Benefits

Components of a PID

Business case

- Why the organization is undertaking the project?
 - The justification for the project
 - Business goals it intends to advance
- State the problem to be solved
 - That could help decision making during the project development
- List expected benefits (and their metrics) such as expected profit or revenue, or required ROI

Success Criteria

- How do we know that the goals are achieved?
 - Primary success factors account for that!
 - These factors are external to the project
- Benefits realization are also secondary success factors
 - They that are usually overlooked and present the bigger risk to the project
 - These might include providing certain value to the customer (e.g. time to market, fault tolerance, satisfaction)

Funding

Components of a PID

Funding amount

- Most projects start out with a funding approved (or estimated)
 - Stating the amount can establish the boundaries of the project and ensure that nobody extrapolates it

Funding status

- Funding often comes with conditions
- These could represent risks for the project funding in the future, such as
 - lack of contingencies for unexpected costs
 - funding not available at once
 - money reserves shared among several projects
 - funding wasn't actually approved to begin with

People and Roles

Components of a PID

Project management team

- Organizational chart or structure
- Description of the roles and responsibilities

Stakeholders

- It is also a good idea to identify all of the stakeholders
 - owners, investors, regulatory bodies, passive stakeholders, and so on
- They can be supporters or opposed
 - Level of support/opposing could vary, which is not often known in the beginning
 - Communication with stakeholders can help clarify that

Processes and Procedures*

Components of a PID

Quality management approach

- Ensure the quality of the project's final products or services according to its customers/end users
- Define standards/criteria and measurements to audit quality
- See verification and validation

Change control approach

- Schedule, budget, team and quality criteria can change mid-project
 - Better plan for changes before they become uncontrollable
 - **When changes occur**, the project manager deals with them according to this

Processes and Procedures*

Components of a PID

Risk management approach

- Identify potential risks, analyze their severity, and plan mitigation/response actions
- More complex projects are extra vulnerable to unseen risks

Communication management approach

- Set up communication channels among the project team and stakeholders
- Important to describe the frequency of communications (see milestones)

Processes and Procedures*

Components of a PID

Project plan

- The core of the initiation document
- Divide the project scope into tasks and assign each task to resources
- Link project resources to schedule and budget
- Provide a timeline upon which the progress can be measured

Project controls

- Actions to ensure the project stays within the constraints
 - See scope, time and budget
 - Tolerance levels should be defined
- How and how often the progress is measured

Class Activity

- Build a Project Initiation Document (PID) for your project idea
- At this stage, you should describe the following:
 - Project Definition / Scope
 - Project Approach and Milestones
 - Business Case
 - Success Criteria (Benefits)
 - Funding Amount and Funding Status
 - Project Management Team
 - *Additional elements of the PID to be developed in future iterations*
- **Tip!** You may use templates from [PMBOK](#) or [PRINCE2](#) as a guide

Chapter 8 – Project Planning: **Understanding the Work**

Breaking down the work

Breaking down the work

- Breakdown of a project into smaller chunks
 - It organizes the team's work into manageable sections
 - Also helps to plan the schedule of the project
- Each component duration is planned in conjunction with its required predecessors and following components
- Two types of breakdown structures:
 - Work Breakdown Structure (WBS) – **process-oriented**, focus on the tasks
 - Product Breakdown Structure (PBS) – **deliverable-oriented**, focus on the outcomes
- Recently, the two approaches are converging

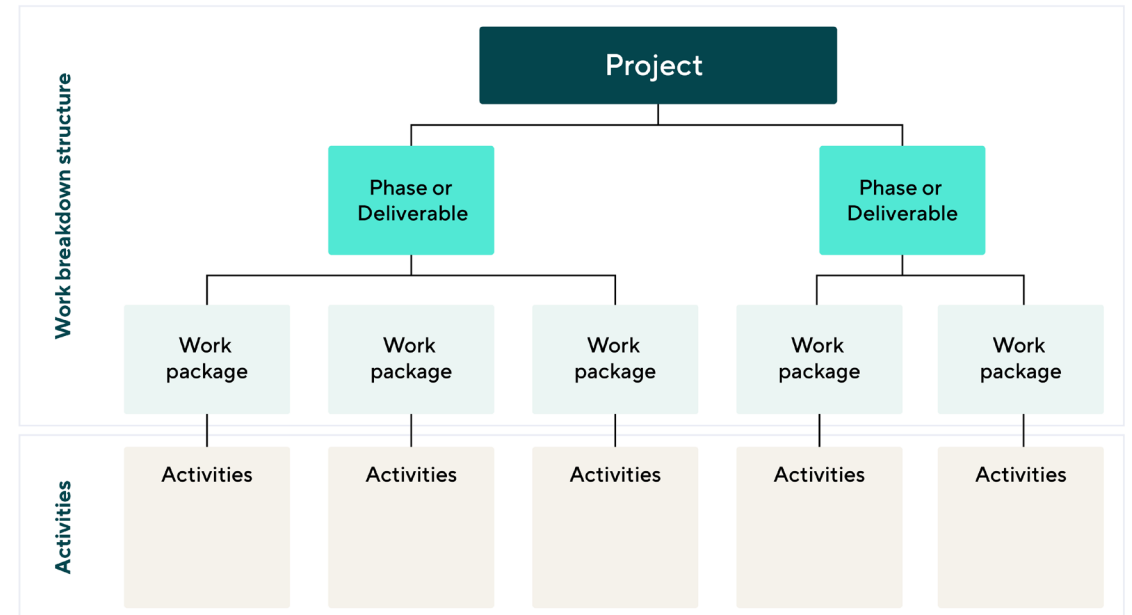
Breaking down the work WBS

WBS of an e-commerce using traditional methods

E-commerce
project

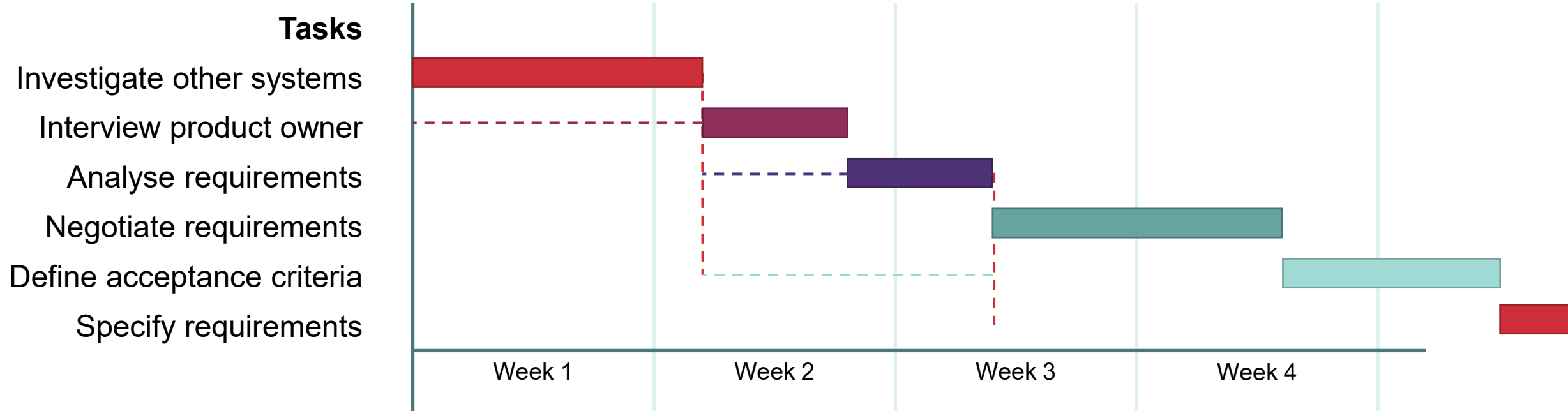
Work packages

- The smallest unit of work that a project can be broken down into when creating the WBS
- Also, a group of related tasks (activities) within a project
 - Tasks are grouped into work packages based on geographical area, discipline, technology, or the time needed to accomplish them
- They look like small projects or sub-projects within a larger project



Sequence of activities

Bar charts or Gantt chart



Breaking down the work

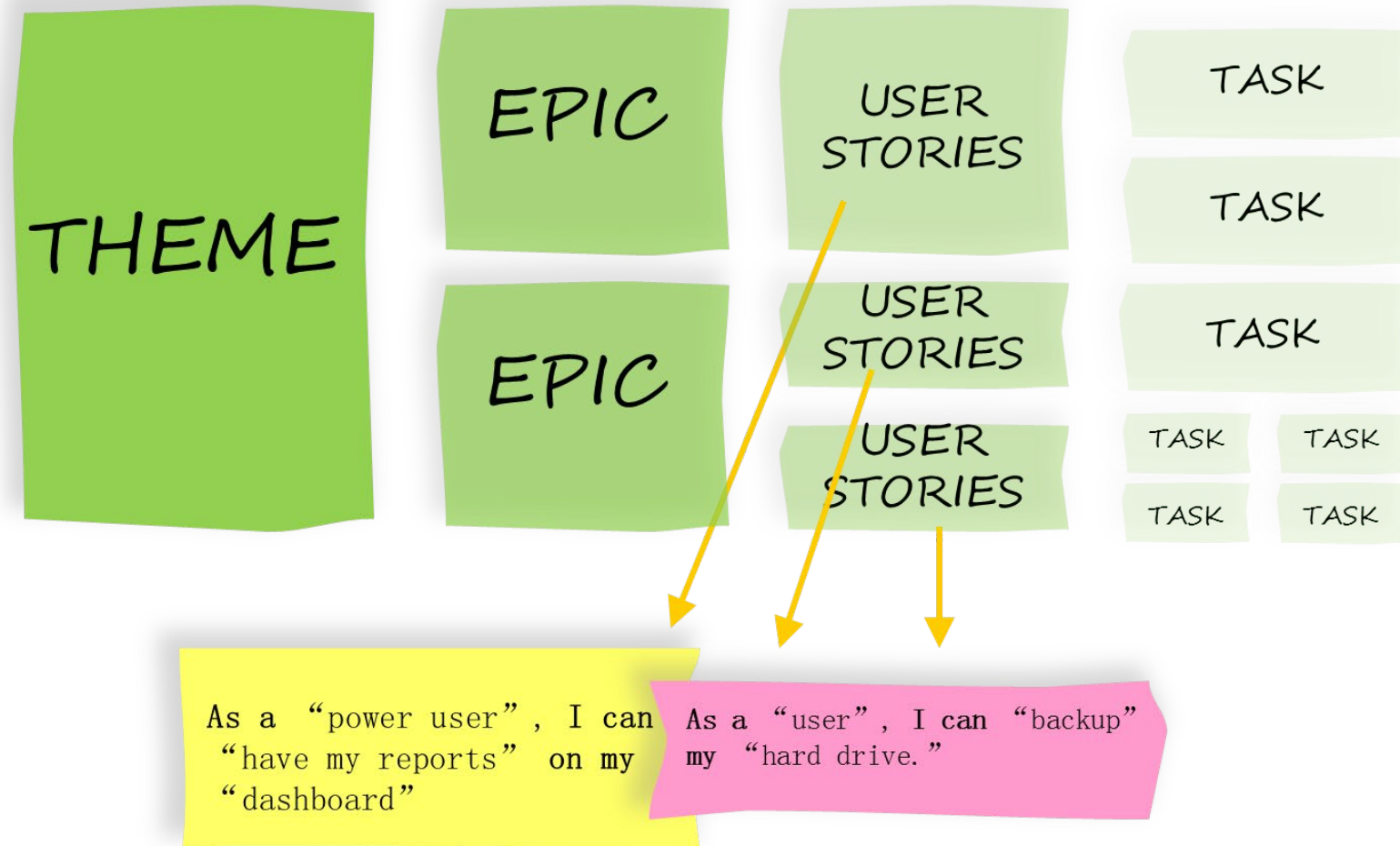
PBS

PBS of a computer machine



Computer

Breaking down the work with Agile



Breakdown structure

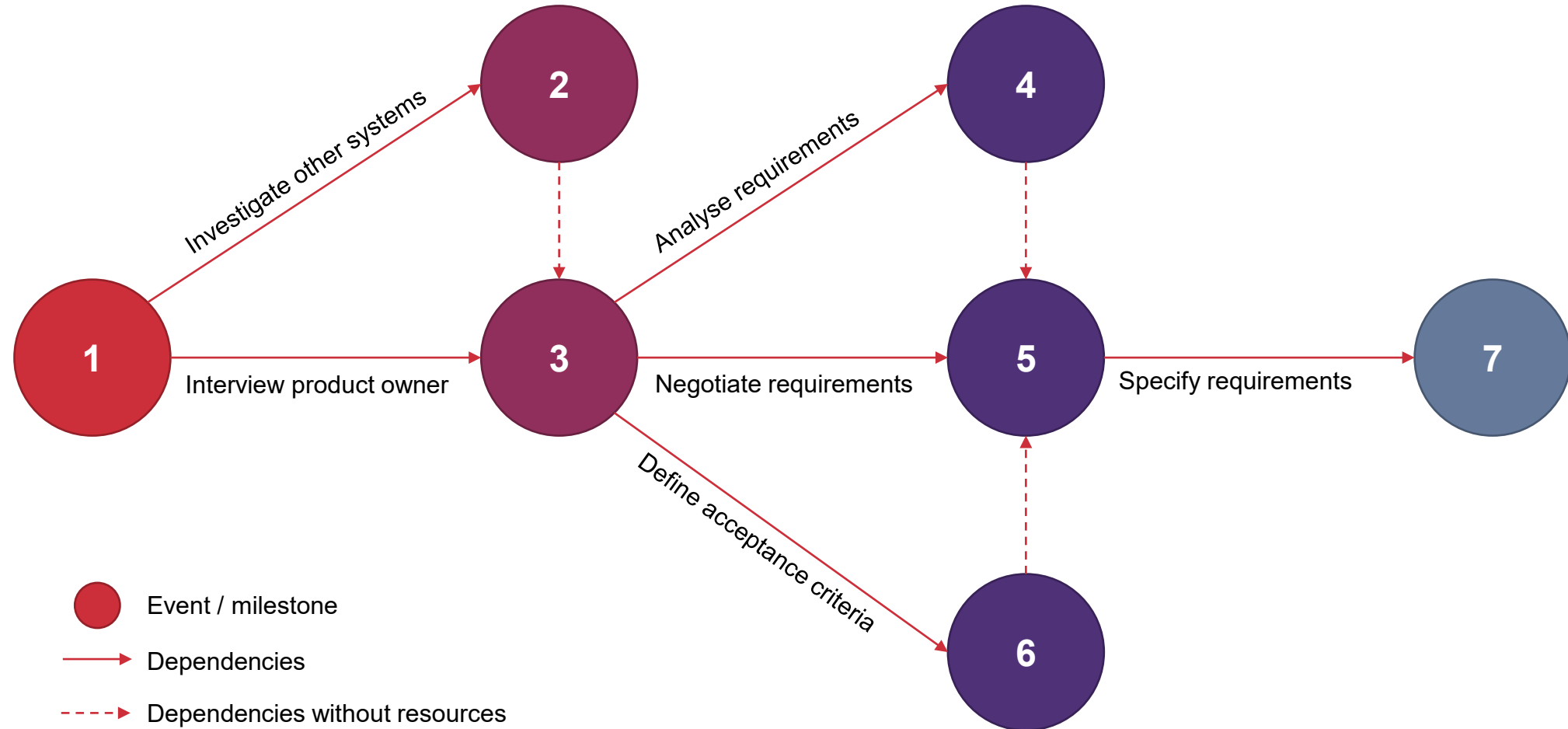
Quick exercise

- Suppose you need to plan a small **weather forecast application**
 - For example Yr.no or Weather.com
- Create a breakdown structure describing the work on the project
 - You can chose either a WBS or PBS
- It should allows for:
 - Forecast (up to a week)
 - Warnings / alerts about the weather
 - Change settings



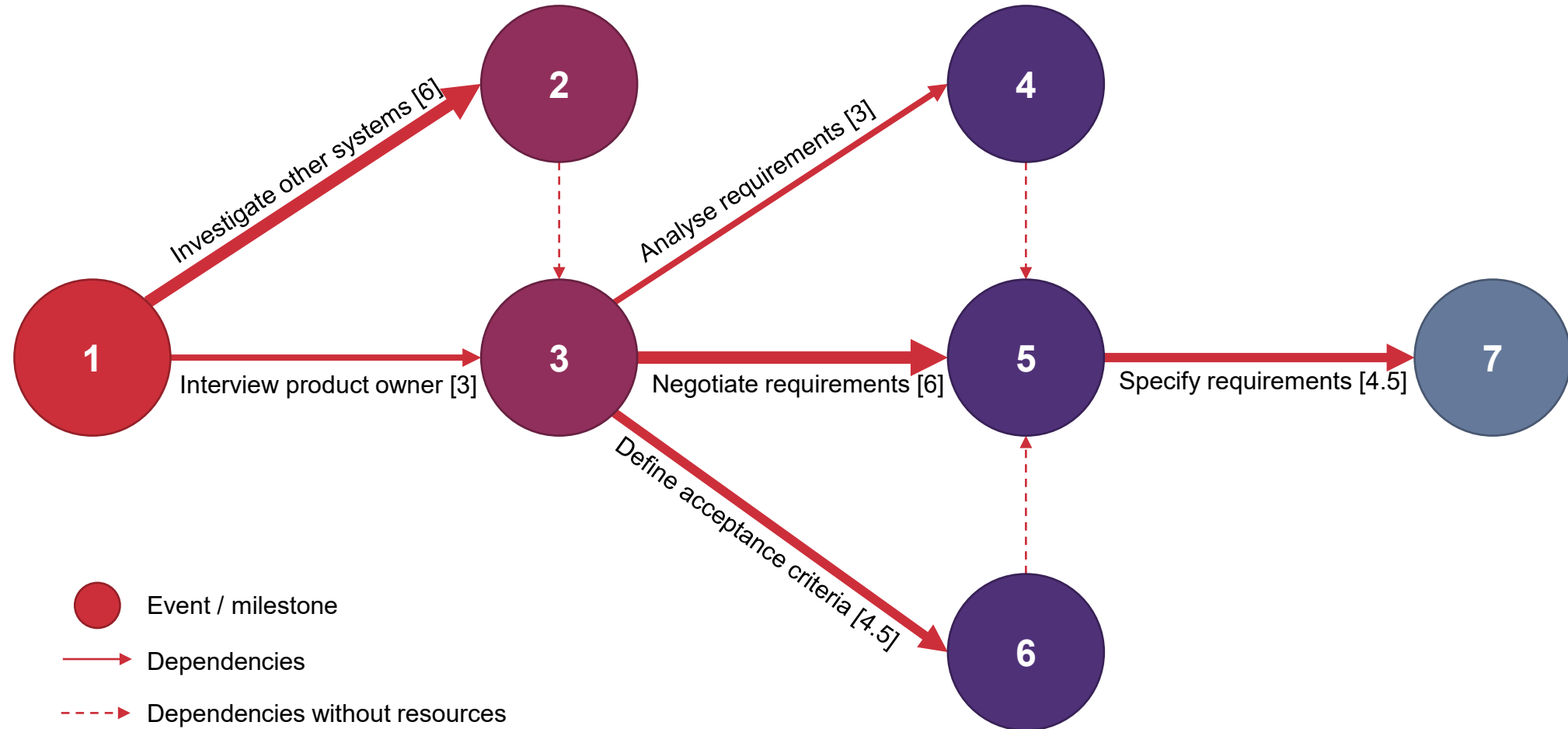
Network diagram (PERT chart)

Highlight dependencies



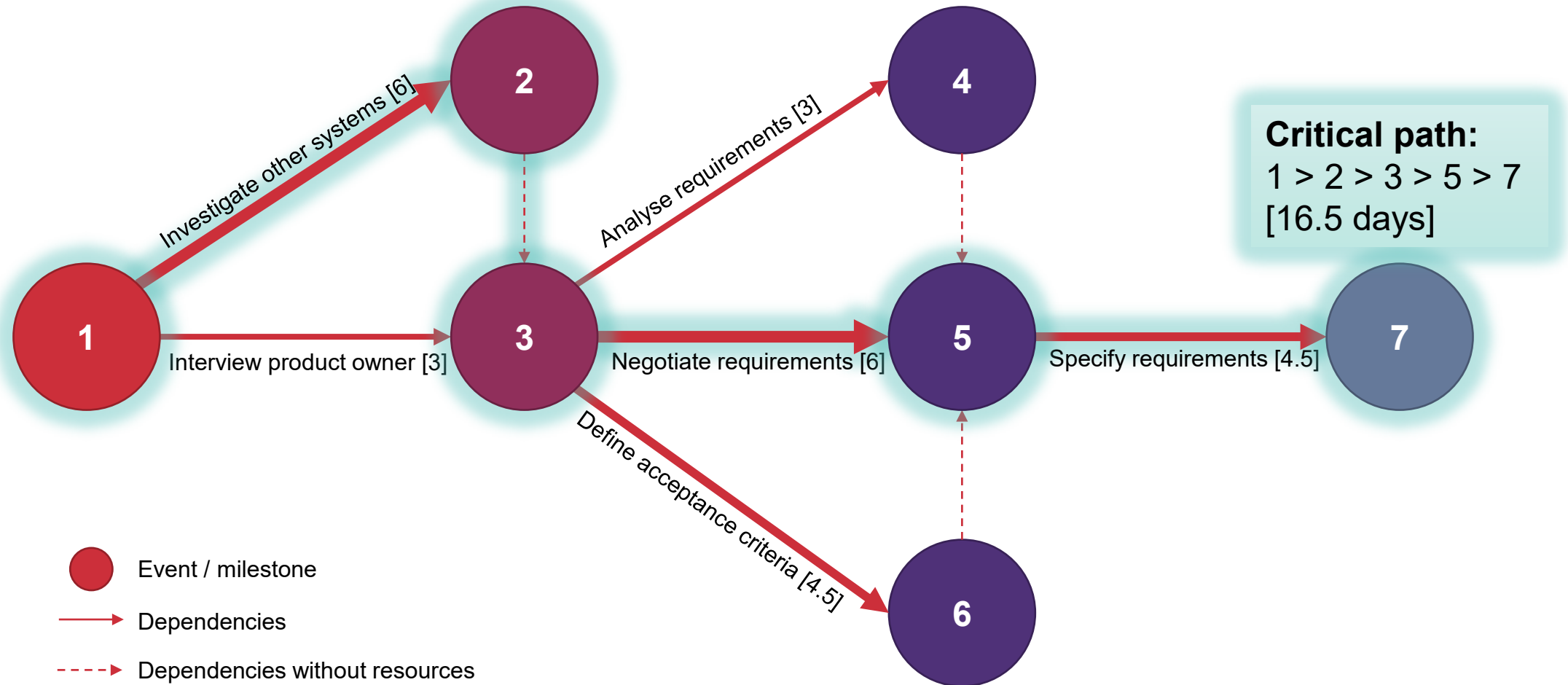
Network diagram (PERT chart)

Calculate LST and LFT



Network diagram (PERT chart)

Identify critical path



Questions?

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Bibliography

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